



GOLI POOJITHA

Roll No.:220150003

B.Tech - Data Science and Artificial Intelligence

Indian Institute of Technology, Guwahati

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GitHub | LinkedIn

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	7.11 (Current)	2022-Present
Senior Secondary	Board of Intermediate Education, A.P	95.9%	2022
Secondary	Board of Secondary Education, A.P	98.5%	2020

EXPERIENCE

- ArisInfra Solutions Limited** May 2025 – Aug. 2025
Software Engineering Intern, Computer Vision for Automated Challan Analysis Bangalore, India
 - Architected a **chromatic extraction workflow** after analyzing over 200 delivery challans, using **RANSAC predicates** for template matching, which data extraction rates from approximately **83% to over 91%**.
 - Replaced the fixed-parameter method with a **locally adaptive model**, implementing a **6-Axis HSV Analysis** to precisely model foreground-background dependencies and generate predictive regression surfaces.
 - Delivered the system using a **pre-computed lookup table** and **weighted distance metric** for OCR image optimization.
- Transfer Learning for Medical Image Analysis** Aug 2025 – ongoing
Bachelor's Thesis Project under Dr.Amulya Kumar Mahato DSAI, IIT Guwahati
 - Validated **transfer learning** by replicating results with CNNs (ResNet, VGGNet), achieving an **80% reduction in training time** and testing domain adaptation methods like **Deep CORAL** for enhanced real-world performance.
 - Conducted a comparative analysis of **3+ generative architectures**, showing **Diffusion Models** improved **structural similarity (SSIM)** over GAN-based methods on a dataset of 1,000+ ultrasound images.

PROJECTS

- Image Captioning with Benchmarking and Robustness Analysis** | *personal Project* Apr 25 | GitHub
 - Developed and benchmarked custom **Vision Transformer + DistilGPT2** model, validating its performance against **SmolVLM** baseline, achieving competitive scores on standard **BLEU, ROUGE, and METEOR** metrics.
 - Assessed model resilience by applying controlled **image perturbations** (upto 80% occlusion) to the test data.
 - Developed a **BERT-based** classifier to distinguish between the generated caption styles of the two models.
 - Achieved **98%** accuracy, proving the models have unique and machine-learnable stylistic fingerprints.
- News Sentiment Analysis** | *Coding Club IITG* June 2024 | GitHub
 - Developed a sentiment analysis model with **TensorFlow** and **Keras**, achieving **95.85%** accuracy, integrated the model into a web application using **Django** and the pickle library for serialization.
 - Harvested and processed news headlines using **Beautiful Soup**, merging with the 'Million News Headlines' dataset.
 - Applied **Vader** and **Snorkel** for sentiment labeling, enhancing model precision by **25%**.
 - Implemented model metrics visualization and a user sentiment feedback for future improvement.
- Magic Eraser** | *Course Project - prof. Chiranjib sur* April 2025 | GitHub
 - Architected an **object removal** model pipeline using **PyTorch**, integrating a fine-tuned **Mask R-CNN** (with a **ResNet-50** backbone) capable of segmenting over **80 object classes** with a two-stage **DeepFillv2** inpainting network.
 - Used **Intersection over Union (IoU)** to accurately select single objects for removal from complex overlapping scenes.
 - Validated effectiveness on **MS-COCO** and real-world photos, producing **3** distinct outputs for review.

TECHNICAL SKILLS

- Programming languages:** C, C++, Python, R
- Web Technologies:** HTML, CSS, Flask, React.js*
- Database:** MySQL, MongoDB
- Data Science:** NumPy, Pandas, Matplotlib, Plotly, TensorFlow, PyTorch, Scikit-learn, PySpark, NLTK
- Miscellaneous:** Git, Quicksight, PowerBI, Tableau

ACHIEVEMENTS

* Elementary proficiency

- CodeForces:** Secured Max rating of **1400+** | **Specialist** | Handle: poojithagoli 2025
 - Achieved a global rank of **2904** in Codeforces Round 1049 (Div. 2) among 25000+ candidates.
- GATE:** Secured **All India Rank (AIR) of 3748** Data Science & AI (DA) paper among 60000+ students 2025
- JEE Mains:** Secured an **All India Rank (AIR) of 3541** among 1.2 million students 2022
- JEE Advanced:** Secured an **All India Rank (AIR) of 5135** among 0.25 million students 2022
- TS EAMCET:** Secured a rank of **234** out of 0.2 million students 2022
- AP EAMCET:** Secured a rank of **274** out of 0.2 million students 2022
- AP Mathematical Olympiad:** Won a **Silver medal** out of 50,000+ students 2019

KEY COURSES TAKEN

- Data Science&Artificial Intelligence** : Deep Learning[†], Machine Learning[†], Data Mining, MultiModal Data Processing, Time Series Analysis, Introduction to AI, Recommendation Systems*, Big Data Analytics[†]
 - Computer Science** : Data Structures and Algorithms[†], Database Management Systems[†], Computer systems, IoT
 - Mathematics:** Linear Algebra, Optimization, Discrete Maths, Applied Probability & Random Processes
- POSITIONS OF RESPONSIBILITY & EXTRACURRICULARS [†]Course has corresponding Lab ^{*}Ongoing Courses

- Events Head, 4i Labs:** Led an event for **project** with students, helping them get **funding**. FY 2024 – 25
- Blog Author:** Authored a post detailing a research paper's methodology and findings. Blog